

High-dimensional Clustering and Signal Recovery under Block Signals

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Abstract

This paper studies computationally efficient methods and their minimax optimality for high-dimensional clustering and signal recovery under block signal structures. We propose two sets of methods, cross-block feature aggregation PCA (CFA-PCA) and moving average PCA (MA-PCA), designed for sparse and dense block signals, respectively. Both methods adaptively utilize block signal structures, applicable to non-Gaussian data with heterogeneous variances and non-diagonal covariance matrices. Specifically, the CFA method utilizes a cross-block statistic to aggregate and select block signals non-parametrically from data with unknown cluster labels. We show that the proposed methods are consistent for both clustering and signal recovery under mild conditions and weaker signal strengths than the existing methods without considering block structures of signals. Furthermore, we derive both statistical and computational minimax lower bounds (SMLB and CMLB) for high-dimensional clustering and signal recovery under block signals, where the CMLBs are restricted to algorithms with polynomial computation complexity. The minimax boundaries partition signals into regions of impossibility and possibility. No algorithm (or no polynomial time algorithm) can achieve consistent clustering or signal recovery if the signals fall into the statistical (or computational) region of impossibility. We show that the proposed CFA-PCA and MA-PCA methods can achieve the CMLBs for the sparse and dense block signal regimes, respectively, indicating the proposed methods are computationally minimax optimal. A tuning parameter selection method is proposed based on post-clustering signal recovery results. Simulation studies are conducted to evaluate the proposed methods. A case study on global temperature change demonstrates their utility in practice.

Keywords: block signals; clustering; high dimensionality; minimax; signal recovery.

1 Introduction

Grouping observations into clusters is an important problem in data analysis. In many scientific studies, especially those involving spatial data, signals that separate different clusters tend to have spatial patterns, so contiguous features share similar mean structures and naturally form blocks of signals. For example, [Chen et al. \(2024\)](#) employed hierarchical clustering on high-resolution satellite data to examine global precipitation patterns, where spatially contiguous regions capture distinct climate patterns. Other examples of data with block signals include medical imaging data, neuroimaging data, copy number variation (CNV) data in genomics, global geoscience data ([Jeng et al., 2010](#); [Feng et al., 2012](#)). Leveraging such block structures enables more accurate clustering and allows detection of weaker signals than methods that treat features as unstructured. In this paper, we develop minimax optimal procedures for clustering and signal recovery for high-dimensional data with block signals under the polynomial-time computational constraint.

There are several recent advances on high-dimensional clustering. [Liu et al. \(2008\)](#) considered detecting clusters in high-dimensional data. [Sun et al. \(2012\)](#) considered regularized k -means clustering. [Cai et al. \(2019\)](#) proposed an expectation-maximization (EM) algorithm for high-dimensional Gaussian mixtures, demonstrating its optimality in convergence rates. Compared to model-based EM algorithms that rely on parametric distribution assumption and k -means method that has a high computational complexity, spectral clustering has gained more attention in high-dimensional data analysis recently. [Löffler et al. \(2021\)](#) established the optimal convergence rate of spectral clustering in Gaussian mixture models with isotropic covariance matrix. [Abbe et al. \(2022\)](#); [Zhang and Zhou \(2024\)](#) developed the leave-one-out analysis framework for eigen-analysis and spectral clustering. Also see spectral clustering for social network data in [Jin \(2015\)](#); [Huang et al. \(2023\)](#). Although

the aforementioned methods were shown effective and optimal in some settings, they didn't consider the phase transition problem of clustering which requires minimax analysis for a finer signal specification. Especially, the problems of clustering and signal recovery for data with block (or spatially contiguous) signals have not been well studied.

This paper adopts a minimax framework to investigate the fundamental limits of clustering in terms of signal strength, sparsity, and block size. [Ingster \(1997\)](#); [Donoho and Jin \(2004\)](#) first derived the detection boundary for testing means, which partitions the parameter space of signal strength and sparsity into the regions of detectable and undetectable. No test can separate the null and alternative hypotheses in the undetectable region. [Kou and Walther \(2022\)](#) extended this result to test block-structured means. [Chen et al. \(2023\)](#); [Qiu and Guo \(2024\)](#) derived the minimax detection boundaries for testing high-dimensional covariance and precision matrices. For clustering, [Jin and Wang \(2016\)](#) introduced the influential feature principal component analysis (IF-PCA) under sparse non-block signals, and subsequently, [Jin et al. \(2017\)](#) established the statistical clustering boundary under non-block signals. However, this statistical boundary may not be attainable by a polynomial-time clustering algorithm in some regions of signals. It is important to understand the limits of clustering algorithms that can be computed in polynomial time.

Some high-dimensional inference tasks are inherently difficult from a computational point of view. The low-degree polynomial conjecture ([Hopkins, 2018](#); [Kunisky et al., 2022](#)) has recently gained attention for its ability to distinguish what is statistically possible from what can be computed in polynomial time. Although not rigorously proven, substantial evidence supporting it was collected in [Hopkins \(2018\)](#). [Löffler et al. \(2022\)](#) used this conjecture to establish the polynomial-time minimax optimal convergence rate for sparse clustering. But, they didn't study the minimax results for signal recovery and neither

considered block signals. [Lyu and Xia \(2023\)](#) applied the conjecture to reveal a statistical-computational gap for estimating low-rank means of two-group Gaussian mixture models.

In this paper, we propose two sets of methods for clustering and signal recovery under sparse and dense block signals. For sparse signals, inspired by IF-PCA ([Jin and Wang, 2016](#)), we first perform feature selection and then cluster the selected features. To handle non-Gaussian distributions and heterogeneous variances, we propose a new method, called the cross-block feature aggregation (CFA) procedure, to aggregate and filter out useful features from clustered data based on pairwise cross-block statistics. The proposed CFA method prevents signal cancellation from unknown cluster labels and avoids estimating unknown variances. It works under non-Gaussian distributions and is more general than IF-PCA designed for the Gaussian distribution with diagonal covariance. After identifying the informative blocks, we perform spectral clustering on the aggregated features of the selected blocks, resulting in the CFA-PCA algorithm. For dense signals, we propose the moving average PCA (MA-PCA) procedure. Different from CFA-PCA, MA-PCA first applies spectral clustering to aggregated features by moving average. Based on the clustering results, we propose a post-clustering signal recovery algorithm for feature selection, inspired by the spatial scan statistics ([Arias-Castro et al., 2005](#); [Jeng et al., 2010](#)). A tuning parameter selection method is proposed based on post-clustering signal recovery results.

We prove that our methods are consistent for both clustering and signal recovery under mild conditions, weaker than the existing methods for sparse clustering and block signal identification. Our results highlight the advantages of the proposed methods in exploiting block structures of signals for both sparse and dense regimes, which are consistent under weaker signal strengths than the existing methods without considering block signals. Simulation studies show that our methods provide more accurate estimates than existing

methods. A case study on global temperature data demonstrates their utility.

Furthermore, we establish both the statistical and computational minimax lower boundaries (SMLB and CMLB) for clustering and signal recovery in a high-dimensional setting with block signals. The SMLB and CMLB partition the parameter space of signal sparsity, strength and block size into the statistical and computational regions of impossibility and possibility, respectively. No algorithm (or no polynomial time algorithm) can achieve consistent clustering or signal recovery if the signals fall into the statistical (or computational) region of impossibility. Compared to [Löffler et al. \(2022\)](#), our results derive the phase transition diagram under block signals, show the gap between SMLB and CMLB, and provide the minimax results for signal recovery. We show that the proposed CFA-PCA and MA-PCA methods can achieve the CMLBs for the sparse and dense block signal regimes, respectively, demonstrating the proposed methods are computationally minimax optimal.

The remainder of this paper is organized as follows. Section 2 presents the problem setting. Sections 3 and 4 introduce the proposed CFA-PCA and MA-PCA procedures. Section 5 derives the SMLB and CMLB, and reveals the computational minimax optimality of the proposed methods. Sections 6 and 7 extend the proposed methods to multi-group clustering and tensor-valued data. Sections 8 and 9 report the simulation and case studies. Additional simulation and proofs are relegated to the Supplementary Material (SM).

2 Problem setting

We consider the two-group high-dimensional clustering setting with block signals. To simplify notations and concentrate on main contributions, we first present our results under vector-valued data, where the variables have a natural ordering in a sequence. Let $\{\mathbf{X}_1, \dots, \mathbf{X}_n\}$ denote a sample of p -dimensional independent observations of size n , where

$\mathbf{X}_i = (X_{i1}, \dots, X_{ip})^\top$. Here, the dimension can be much larger than the sample size. Following the high-dimensional clustering model proposed in Jin et al. (2017), we assume the cluster label of each observation is independent and identically distributed (IID) from a Bernoulli distribution. Let $\ell_i \sim 2\text{Bernoulli}(\varpi/(1+\varpi)) - 1$, where ϖ is a positive constant indicating the proportion of the two groups. Let $\boldsymbol{\mu}_1 = \mathbb{E}(\mathbf{X}_i | \ell_i = 1)$ and $\boldsymbol{\mu}_{-1} = \mathbb{E}(\mathbf{X}_i | \ell_i = -1)$ be the means of the two groups. We consider the following model for the observation data:

$$\mathbf{X}_i = \boldsymbol{\mu}_{\ell_i} + \mathbf{Z}_i, \quad \mathbf{Z}_i \stackrel{\text{IID}}{\sim} F \text{ with mean zero and covariance } \boldsymbol{\Sigma}, \quad i = 1, \dots, n, \quad (2.1)$$

where $\boldsymbol{\delta} = (\delta_1, \dots, \delta_p)^\top = \boldsymbol{\mu}_1 - \boldsymbol{\mu}_{-1}$ is the mean contrast and $\boldsymbol{\Sigma} = (\sigma_{j_1 j_2})$ is unknown. We investigate both clustering and signal recovery problems, which aim to identify the labels $\boldsymbol{\ell} = (\ell_1, \dots, \ell_n)^\top$ and to recover the signal support $S(\boldsymbol{\delta}) = \{j : \delta_j \neq 0\}$, respectively.

We consider a block structure for the contrast mean $\boldsymbol{\delta}$. Its nonzero components are distributed across non-overlapping ordered blocks $\mathbb{I}_0 = \{I_{0,g} : 1 \leq g \leq m\}$, where $I_{0,g} = \{j_g, j_g + 1, \dots, j_g + l_g - 1\}$ and $j_g + l_g \leq j_{g+1}$ for all g , l_g denotes the length of g th block, and m denotes the number of blocks. Let $d_0 = \min_g \{j_{g+1} - (j_g + l_g - 1)\}$ be the minimum distance between blocks. The number of nonzero entries of $\boldsymbol{\delta}$ is denoted as $s = |S(\boldsymbol{\delta})| = \sum_{g=1}^m l_g$, where $|A|$ denotes the cardinality of a set A . We summarize the key notations used in the proposed methods in Table S.7 of the SM.

In this paper, we consider the setting of expanding signal numbers that $s \rightarrow \infty$, and impose the following conditions on the block signal strength and the data distribution.

Assumption 1 For any $\mathbf{t} \in \mathbb{R}^p$ and a constant $C > 0$, $\mathbb{E}\{\exp(\mathbf{t}^\top \mathbf{Z}_i)\} \leq \exp(C\mathbf{t}^\top \boldsymbol{\Sigma} \mathbf{t}/2)$.

Assumption 2 The minimum distance between blocks satisfies $d_0 > \log(p) \cdot \max_{1 \leq g \leq m} |I_{0,g}|$.

Assumption 3 For each $I_{0,g} \in \mathbb{I}_0$, there exists $\delta_{0,g} > 0$ and $c_0 \geq 1$ such that $c_0^{-1} \delta_{0,g} \leq |\delta_j| \leq c_0 \delta_{0,g}$ and $\text{sign}(\delta_j) = e_g$ for $j \in I_{0,g}$ and $e_g = \pm 1$.

Assumption 1 assumes the data are sub-Gaussian distributed, which is commonly made in high-dimensional literature (Bickel and Levina, 2008; Cai et al., 2010, 2023). Assumption 2 assumes that all blocks of signals are asymptotically separable, meaning the distance between any pair of signal blocks is much larger than their sizes. This assumption is needed for the correct recovery of each block. Intuitively, if the distance between two signal blocks is much smaller than their sizes, they can be viewed as one block asymptotically. A similar condition was made in Jeng et al. (2010) for signal identification under IID data without unknown clusters. If the left ends of all blocks, $\{j_1, \dots, j_m\}$, are randomly selected from $\{1, \dots, p\}$, we show in Lemma S1 in the SM that

$$d_0 \geq 2pm^{-2} \log^{-1}(p) \quad (2.2)$$

with probability converging to 1 if $m = o((p/l_{\max})^{1/2} \log^{-1/2}(p))$, where $l_{\max} = \max_{1 \leq g \leq m} l_g$. It follows that $d_0 \gg l_{\max}$, and Assumption 2 is satisfied in the sparse block signal regime. Assumption 3 assumes the nonzero mean differences are at the same order and have the same sign within each block, but they can vary across blocks. Such block signal structures are commonly observed in geoscience and medical imaging data. Kou and Walther (2022) considered a similar block signal setting for testing means. However, they set equal signal for all dimensions, which corresponds to $c_0 = 1$ and $\delta_{0,g} \equiv \delta_0$ in Assumption 3.

Under Assumption 1–3, we allow heterogeneous block lengths $\{l_g\}_{g=1}^m$ and block signal rates $\{\delta_{0,g}\}_{g=1}^m$. This accommodates non-block signals, and blocks with fixed and diverging lengths as $p \rightarrow \infty$. For example, $l_g \asymp p^{\alpha_g}$ for $\alpha_g \in [0, 1)$, where $a_{1,n} \asymp a_{2,n}$ denotes two positive sequences $\{a_{1,n}\}$ and $\{a_{2,n}\}$ being at the same order such that $c_1 \leq a_{1,n}/a_{2,n} \leq c_2$ for all n and constants $c_1, c_2 > 0$. Let $\|\cdot\|$ denote the vector Euclidean norms. Let $\mu_{0,g}^2 = l_g \delta_{0,g}^2$ be the accumulated signal strength of the g th block. Let

$$\bar{\mu}_0^2 = \frac{1}{m} \sum_{g=1}^m \mu_{0,g}^2 \asymp \frac{\|\delta\|^2}{m} \quad \text{and} \quad l_a = \min\{l_g : \mu_{0,g} > \bar{\mu}_0\} \quad (2.3)$$

be the average accumulated signal strength and the minimum block length with $\mu_{0,g} > \bar{\mu}_0$. Depending on the value of m , the setting can be categorized as sparse and dense block signal regimes, which require different clustering algorithms as introduced in Sections 3 and 4. A rough criterion is $m \ll \sqrt{p/l_a}$ and $m \gg \sqrt{p/l_a}$ as explained after Theorem 3, which is similar to the separation $\sqrt{p}$ for non-block signals (Donoho and Jin, 2004; Jin et al., 2017). Under the setting for minimax lower bounds in Section 5, there exists a clear transition between the sparse and dense block signals as revealed by Corollary 1.

Our setting and assumptions for high-dimensional clustering and signal recovery under block signals are relatively general. Most of the existing high-dimensional clustering methods with sparsity constraints require the Gaussianity assumption and don't consider block signals (Jin and Wang, 2016; Cai et al., 2019; Löffler et al., 2022). Jeng et al. (2010); Kou and Walther (2022) also assumed Gaussian data with identity covariance for block signal detection and identification. In contrast, our setting relaxes those assumptions, and the proposed algorithms in Sections 3 and 4 can be extended to multi-group clustering and tensor data, making them applicable to geoscience applications.

3 Methods for sparse block signals

In this section, we introduce the polynomial-time algorithms for clustering and signal recovery under sparse block signals. The minimax results in Section 5 show that consistent clustering requires stronger signals than signal recovery for signals with small m , and vice versa for signals with large m . This indicates that we should conduct pre-clustering signal recovery first and cluster the observations based on the selected features for sparse block signals. Instead, for dense block signals, clustering should be applied first, followed by a post-clustering signal recovery procedure.

Let $\boldsymbol{\mu} = (\varpi\boldsymbol{\mu}_1 + \boldsymbol{\mu}_{-1})/(1 + \varpi)$ be the overall mean of $\mathbf{X}_i$ under the model in (2.1), and $\bar{\mathbf{X}} = \sum_{i=1}^n \mathbf{X}_i/n$ be its estimate. Let $\mathbf{X}_i^* = \mathbf{X}_i - \boldsymbol{\mu}$ and $\tilde{\mathbf{X}}_i = \mathbf{X}_i - \bar{\mathbf{X}}$. We use the centered data $\{\tilde{\mathbf{X}}_i\}$ for clustering. Note that the model in (2.1) can be expressed as $\mathbf{X}_i^* = \tilde{\ell}_i \boldsymbol{\delta}/(1 + \varpi) + \mathbf{Z}_i$, where $\tilde{\ell}_i = 1$ for $\ell_i = 1$ and $\tilde{\ell}_i = -\varpi$ for $\ell_i = -1$. Sparse feature selection for clustered data is difficult. The χ^2 statistic $n^{-1/2} \sum_{i=1}^n \tilde{X}_{ij}^2$ cannot identify the nonzero δ_j if σ_{jj} is unknown. To solve this problem, Jin and Wang (2016) used Kolmogorov-Smirnov statistic for feature screening under Gaussian mixture distributions with diagonal covariance. We propose a new feature selection method for clustered data under block signals and non-Gaussian distributions with bandable covariance matrices, defined in (3.1).

To remove the unknown variance in the χ^2 statistic, we introduce a pairwise screening procedure. Let $\mathbb{J}_p(h_1) = \{(j, \dots, j+h) : 1 \leq h \leq h_1, 1 \leq j < j+h \leq p\}$ be the collection of all blocks with length no larger than h_1 , where $|\mathbb{J}_p(h_1)| \leq ph_1$. For $I_1, I_2 \in \mathbb{J}_p(h_1)$, consider the cross-product $W_i(I_1, I_2) = \tilde{X}_i(I_1)\tilde{X}_i(I_2)$, where $\tilde{X}_i(I) = \sum_{j \in I} \tilde{X}_{ij}/\sqrt{|I|}$. Let $Z_i(I) = \sum_{j \in I} Z_{ij}/\sqrt{|I|}$, $\delta(I) = \sum_{j \in I} \delta_j/\sqrt{|I|}$, and $\sigma(I_1, I_2) = \mathbb{E}\{Z_i(I_1)Z_i(I_2)\}$. Since $\mathbb{E}\{X_i^*(I_1)X_i^*(I_2)\} = \tilde{\ell}_i^2 \delta(I_1)\delta(I_2)/(1 + \varpi)^2 + \sigma(I_1, I_2)$ where $X_i^*(I) = \sum_{j \in I} X_{ij}^*/\sqrt{|I|}$, the signals in $\{W_i(I_1, I_2)\}_{i=1}^n$ would not be canceled by taking average, and those cross-products do not involve the variances of $Z_i(I)$. If $\sigma(I_1, I_2)$ is small, we can use $\bar{W}(I_1, I_2) = n^{-1/2} \sum_{i=1}^n W_i(I_1, I_2)$ to test for $\delta(I_1)\delta(I_2) = 0$, and hence, to identify the blocks with nonzero means. This shows the advantages of using cross-products for signal screening.

For data with a natural ordering or collected in spatial locations such as those in earth science problems, the dependence between two variables decreases as their distance increases. The following “bandable” class (Bickel and Levina, 2008; Cai et al., 2010) describes such covariance matrices for variables observed in a sequence:

$$\mathfrak{U}(\gamma, C) = \left\{ \boldsymbol{\Sigma} = (\sigma_{j_1 j_2}) : \max_{|j_1 - j_2| > k} |\sigma_{j_1 j_2}| \leq Ck^{-(\gamma+1)} \text{ for all } k > 0 \right\} \quad (3.1)$$

for $\gamma > 0$ and $C > 0$. This bandable class can be extended to spatial data that encompasses a wide range of spatial covariance functions (Sun et al., 2024). Under the class in (3.1), $\sigma(I_1, I_2) \rightarrow 0$ as the distance between I_1 and I_2 diverges. Note that γ can be treated as ∞ if Σ is diagonal. Therefore, we consider the pair of blocks that are far apart.

Let $\mathcal{E}_p(\{j, \dots, j + \tilde{c}_1\}, \tilde{c}_2) = \{j - \tilde{c}_2, \dots, j + \tilde{c}_1 + \tilde{c}_2\} \cap \{1, \dots, p\}$ be the expanding operator that expands the interval $\{j, \dots, j + \tilde{c}_1\}$ from both ends by a window $\tilde{c}_2$, where $\tilde{c}_1, \tilde{c}_2 > 0$. For $I_1 \in \mathbb{J}_p(h_1)$ and a window size $h_2 > 0$, we consider its cross products with blocks that are at least h_2 apart. Let

$$\tilde{I}_1 = \operatorname{argmax}_{I_2 \in \mathbb{J}_p(h_1), I_2 \cap \mathcal{E}_p(I_1, h_2) = \emptyset} |\overline{W}(I_1, I_2)| \quad \text{and} \quad W_0(I_1) = \overline{W}(I_1, \tilde{I}_1) \quad (3.2)$$

be the block having the largest cross-product with I_1 that satisfies $\tilde{I}_1 \cap \mathcal{E}_p(I_1, h_2) = \emptyset$. We use $W_0(I)$ to detect signals in the block I . Let $\hat{\sigma}_w^2(I) = n^{-1} \{\sum_{i=1}^n W_i^2(I, \tilde{I}) - W_0^2(I)\}$ be the sample variance of $\{W_i(I, \tilde{I})\}_{i=1}^n$. We identify the significant blocks if their standardized aggregated average $|W_0(I)|/\hat{\sigma}_w(I)$ is larger than the threshold $t_{w,p} = \sqrt{6 \log(ph_1)}$. Let $\mathbb{Q}_w^{(1)} = \{I \in \mathbb{J}_p(h_1) : |W_0(I)|/\hat{\sigma}_w(I) > t_{w,p}\}$. We select the first identified block as the largest $|W_0(I)|$ from $\mathbb{Q}_w^{(1)}$, denoted as $\hat{I}_1 = \operatorname{argmax}_{I \in \mathbb{Q}_w^{(1)}} |W_0(I)|$, and search the next significant block in the set $\mathbb{Q}_w^{(2)} = \mathbb{Q}_w^{(1)} \setminus \{I \in \mathbb{Q}_w^{(1)} : I \cap \mathcal{E}_p(\hat{I}_1, \lfloor h_1/2 \rfloor) \neq \emptyset\}$ by removing any block intersecting an expanded block of $\hat{I}_1$ with length $|\hat{I}_1| + h_1$ from $\mathbb{Q}_w^{(1)}$. We continue this procedure until $\mathbb{Q}_w^{(k)}$ becomes empty. The requirement $I \cap \mathcal{E}_p(\hat{I}_k, \lfloor h_1/2 \rfloor) = \emptyset$ in each step ensures that all identified blocks are at least $h_1/2$ separated from each other, guaranteeing the consistency of the number of identified blocks. We call this procedure the cross-block feature aggregation (CFA) method for pre-clustering signal identification.

Let $\hat{\mathbb{I}}_{\text{cfa}} = \{\hat{I}_1, \hat{I}_2, \dots, \hat{I}_{\hat{m}}\}$ denote the $\hat{m}$ identified blocks by the proposed CFA method. Then, we perform clustering solely on the aggregated mean of each selected block. Let $\tilde{\mathbf{X}} = (\tilde{\mathbf{X}}_1, \dots, \tilde{\mathbf{X}}_n)^T = (\tilde{\mathbf{X}}_{(1)}, \dots, \tilde{\mathbf{X}}_{(p)})$ denote the centered data, where $\tilde{\mathbf{X}}_{(j)} = (\tilde{X}_{1j}, \dots, \tilde{X}_{nj})^T$.

Algorithm 1 CFA block identification

Calculate $W_0(I)$, $\hat{\sigma}_w(I)$ for $\forall I \in \mathbb{J}_p(h_1)$ via (3.2);

$\mathbb{Q}_w^{(1)} \leftarrow \{I \in \mathbb{J}_p(h_1) : |W_0(I)|/\hat{\sigma}_w(I) > t_{w,p}\}$;

For $k = 1, 2, \dots$, **while** $\mathbb{Q}_w^{(k)} \neq \emptyset$:

$\hat{I}_k \leftarrow \operatorname{argmax}_{I \in \mathbb{Q}_w^{(k)}} |W_0(I)|$;

$\mathbb{Q}_w^{(k+1)} \leftarrow \mathbb{Q}_w^{(k)} \setminus \{I : I \cap \mathcal{E}_p(\hat{I}_k, \lfloor h_1/2 \rfloor) \neq \emptyset\}$;

Return: $\hat{m} \leftarrow k$ and $\hat{\mathbb{I}}_{\text{cfa}} \leftarrow \{\hat{I}_1, \dots, \hat{I}_{\hat{m}}\}$.

Algorithm 2 CFA-PCA clustering

$\hat{\mathbf{Y}}_{(k)}^{(\text{cfa})} \leftarrow |\hat{I}_k|^{-1/2} \sum_{j \in \hat{I}_k} \tilde{\mathbf{X}}_{(j)}$ for $k = 1, \dots, \hat{m}$;

$\hat{\mathbf{Y}}^{(\text{cfa})} \leftarrow (\hat{\mathbf{Y}}_{(1)}^{(\text{cfa})}, \dots, \hat{\mathbf{Y}}_{(\hat{m})}^{(\text{cfa})})$;

$\hat{\boldsymbol{\xi}}_{\text{cfa}} \leftarrow \text{leading eigenvector of } \hat{\mathbf{Y}}^{(\text{cfa})} \hat{\mathbf{Y}}^{(\text{cfa})\top}$;

Return: $\hat{\boldsymbol{\ell}}_{\text{cfa}} \leftarrow \text{sign}(\hat{\boldsymbol{\xi}}_{\text{cfa}})$.

Let $\hat{\mathbf{Y}}^{(\text{cfa})} = (\hat{\mathbf{Y}}_{(1)}^{(\text{cfa})}, \dots, \hat{\mathbf{Y}}_{(\hat{m})}^{(\text{cfa})})$ be the aggregated data matrix of the selected features, where $\hat{\mathbf{Y}}_{(k)}^{(\text{cfa})} = |\hat{I}_k|^{-1/2} \sum_{j \in \hat{I}_k} \tilde{\mathbf{X}}_{(j)}$ for $k = 1, \dots, \hat{m}$. The cluster labels $\boldsymbol{\ell}$ are estimated by

$$\hat{\boldsymbol{\ell}}_{\text{cfa}} = \text{sign}(\hat{\boldsymbol{\xi}}_{\text{cfa}}), \quad (3.3)$$

where $\hat{\boldsymbol{\xi}}_{\text{cfa}}$ is the first eigenvector of $\hat{\mathbf{Y}}^{(\text{cfa})} \hat{\mathbf{Y}}^{(\text{cfa})\top}$. We call this clustering method cross-block feature aggregation PCA (CFA-PCA). Algorithms 1 and 2 outline its main steps. The computation cost of CFA-PCA is $O(n(ph_1)^2 + n^2m)$. Note that for diagonal covariance matrices, the separation window size h_2 in (3.2) can be chosen as 0, and the CFA-PCA procedure can be simplified. Section S3.1 of the SM presents such a simplified CFA-PCA procedure without the hyperparameter h_2 under a diagonal $\boldsymbol{\Sigma}$.

Let C denote a positive constant which may change from case to case. Let $\|\cdot\|_2$ denote the matrix L_2 norm, and $\|\cdot\|_\infty$ denote the vector maximum norm. The following theorem shows the consistency of $\hat{\boldsymbol{\ell}}_{\text{cfa}}$ under a suitable condition on $\|\boldsymbol{\delta}\|$.

Theorem 1 *Under Assumptions 1-3, $\boldsymbol{\Sigma} \in \mathfrak{U}(\gamma, C)$ in (3.1) and $\max\{\|\boldsymbol{\Sigma}\|_2, \|\boldsymbol{\Sigma}^{-1}\|_2\} \leq C$, if $\sqrt{n}l_{\max}d_0^{-\gamma-1} = o(1)$, $h_1 = c_1l_{\max}$ and $h_2 = c_2d_0$ for any constants $c_1 \geq 1$ and $c_2 \in (0, 1)$, $\log(pl_{\max}) = o(n^{1/4})$, $\boldsymbol{\delta} = \boldsymbol{\mu}_1 - \boldsymbol{\mu}_{-1}$ satisfies the signal-to-noise ratio condition*

$$(\sqrt{n}/m)\|\boldsymbol{\delta}\|^2 > c_3 \log p, \quad \min\{(n/m)^{1/2}, 1\}\|\boldsymbol{\delta}\|^2 > c_3 \log n \quad (3.4)$$

for $c_3 > 0$, and there exist at least two blocks satisfying $\mu_{0,g} > \bar{\mu}_0$, then we have $\min\{\|\hat{\boldsymbol{\ell}}_{\text{cfa}} + \boldsymbol{\ell}\|_\infty, \|\hat{\boldsymbol{\ell}}_{\text{cfa}} - \boldsymbol{\ell}\|_\infty\} \rightarrow 0$ with probability approaching 1 as $n, p \rightarrow \infty$.

Theorem 1 establishes the consistency of the proposed CFA-PCA procedure for clustering. The first signal-to-noise condition in (3.4) is for block feature selection, which ensures $\sqrt{n}\bar{\mu}_0^2 \rightarrow \infty$ so that the blocks with accumulated signal strength $\mu_{0,g}$ larger than $\bar{\mu}_0$ can be identified. The second signal-to-noise condition on $\min\{(n/m)^{1/2}, 1\}$ in (3.4) is much weaker than the classical condition $\min\{(n/p)^{1/2}, 1\}$ for spectral clustering without feature selection (Cai and Zhang, 2018; Abbe et al., 2022). Our CFA procedure reduces the effective dimension from p to m for clustering. Conditions $h_2 = c_2 d_0$ and $\sqrt{n}l_{\max}d_0^{-\gamma-1} = o(1)$ together ensure $\sqrt{n}\sigma(I_1, I_2) = o(1)$ for any pair of blocks separated by at least d_0 , which prevents false signals due to variable correlations in the CFA procedure. Under (2.2), $n \leq Cp$, $m \leq C$ and $l_g \asymp p^\alpha$ for $\alpha \in [0, 1)$, a sufficient condition for $\sqrt{n}l_{\max}d_0^{-\gamma-1} = o(1)$ is $\gamma > \alpha - 1/2$, which places no restriction on γ when $\alpha < 1/2$. We provide a data-driven tuning parameter selection method for h_1 and h_2 in the simulation section.

Löffler et al. (2022) proposed a sparse spectral clustering method for non-block signals under Gaussian data with identity covariance, which is based on sparse PCA (Zou and Xue, 2018). Their method is consistent under an almost equivalent signal-to-noise ratio condition, as in (3.4), for non-block signals, and a slightly weaker dimensionality condition, $\log p \leq n$. For non-block signals and diagonal covariance matrices, it turns out the proposed method does not require Assumption 2; see Theorem S1 in the SM. Therefore, under this simpler setting, our method can handle more general distributions with unknown variances.

The following theorem shows that the selected blocks $\hat{\mathbb{I}}_{\text{cfa}}$ by the proposed CFA method is consistent with the true set of signals $\mathbb{I}_0$, if the signal strengths within each block are the same. Let $D(I_1, I_2) = 1 - |I_1 \cap I_2|(|I_1||I_2|)^{-1/2}$ be the dissimilarity between sets I_1 and I_2 .

Theorem 2 *Under Assumptions 1 and 2, Assumption 3 with $c_0 = 1$, $\Sigma \in \mathfrak{U}(\gamma, C)$ in (3.1) and $\max\{\|\Sigma\|_2, \|\Sigma^{-1}\|_2\} \leq C$, if $\sqrt{n}l_{\max}d_0^{-\gamma-1} = o(1)$, $h_1 = c_1 l_{\max}$ and $h_2 =$*

$c_2 d_0$ for any constants $c_1 \geq 1$ and $c_2 \in (0, 1)$, $\log(pl_{\max}) = o(n^{1/4})$, and δ satisfies $\sqrt{n} \min_{1 \leq g \leq m} \mu_{0,g}^2 > c_3 \log p$ for a positive constant c_3 , then we have that $\mathbb{P}(\hat{m} = m) \rightarrow 1$ and $\mathbb{P}(\max_{I \in \mathbb{I}_0} \min_{\hat{I} \in \hat{\mathbb{I}}_{\text{cfa}}} D(\hat{I}, I) > \zeta) \rightarrow 0$ for any $\zeta > 0$.

Compared to Theorem 1, a perfect signal recovery requires an additional condition that the signal strengths in each block $I_{0,g}$ are same. Otherwise, the aggregated signal $\delta^2(I_{0,g})$ may not be the largest among $\{\delta^2(I) : I \subseteq I_{0,g}\}$. The signal recovery procedure would select the subset that maximizes $\delta^2(I)$ instead of $I_{0,g}$, despite it suffices for successful clustering as discussed in Theorem 1. The same equal strength condition is also required in Arias-Castro et al. (2005); Jeng et al. (2010) for block signal identification.

4 Methods for dense block signals

In this section, we consider clustering and signal recovery for dense block signals. Different from the sparse signal regime, we first cluster the observations and then identify the components with signals based on the estimated clusters. To utilize the block signal structure, we conduct clustering on the aggregated variables by moving average with window size h_3 . Let $\mathbf{Y}_{(g)}^{(\text{ma})} = \frac{1}{\sqrt{h_3}} \sum_{j=g}^{g+h_3-1} \tilde{\mathbf{X}}_{(j)}$ and $\mathbf{Y}^{(\text{ma})} = (\mathbf{Y}_{(1)}^{(\text{ma})}, \dots, \mathbf{Y}_{(p-h_3+1)}^{(\text{ma})})$ be the moving average matrix of $\tilde{\mathbf{X}}$, where h_3 is a tuning parameter. We estimate the cluster labels ℓ by

$$\hat{\ell}_{\text{ma}} = \text{sign}(\hat{\boldsymbol{\xi}}_{\text{ma}}), \quad (4.1)$$

where $\hat{\boldsymbol{\xi}}_{\text{ma}}$ is the first eigenvector of the matrix $\mathbf{Y}^{(\text{ma})} \mathbf{Y}^{(\text{ma})\text{T}}$. We call this method moving average PCA (MA-PCA) for clustering. Its computation cost is $O(np(h_3 + n))$. The following theorem establishes its consistency.

Theorem 3 Under Assumptions 1-3, for a constant $c_4 > 0$, if δ and h_3 satisfy

$$\min \{ \sqrt{n/(ph_3)}, h_3^{-1} \} \|\boldsymbol{\Sigma}\|_2^{-1} \sum_{g=1}^m \min \{ h_3, l_g \} \mu_{0,g}^2 > c_4 \log n, \quad (4.2)$$

we have $\min\{\|\hat{\ell}_{\text{ma}} + \ell\|_\infty, \|\hat{\ell}_{\text{ma}} - \ell\|_\infty\} \rightarrow 0$ with probability approaching 1 as $n, p \rightarrow \infty$.

Theorem 3 demonstrates that the proposed MA-PCA method can consistently cluster the observations under general covariances if the signal-to-noise ratio satisfies (4.2). This condition is similar to that in Abbe et al. (2022) for non-block signals by taking $h_3 = l_g = 1$. In the block signal setting, the condition on the signal strength $\|\delta\|^2$ can be weakened by a properly chosen h_3 . Compared to CFA-PCA in Theorem 1, if $\|\Sigma\|_2 \leq C$ and $n \leq Cp$, by taking $h_3 \asymp l_a$, sufficient conditions to (3.4) and (4.2) can be simplified as $\min\{\sqrt{nm}^{-1}, 1\}\|\delta\|^2 > c \log p$ and $\min\{\sqrt{n}(p/l_a)^{-1/2}, 1\}\|\delta\|^2 > c \log p$ for CFA-PCA and MA-PCA, respectively. When $m \ll \sqrt{p/l_a}$ meaning the sparse block signal regime, CFA-PCA requires a weaker condition on $\|\delta\|$, whereas for the dense block signal regime that $m \gg \sqrt{p/l_a}$, MA-PCA requires a weaker condition on $\|\delta\|$. This indicates the necessity of designing different algorithms for different sparsity regimes. Section 5 shows that both methods achieve computational minimax optimality in their respective sparsity regimes. A data-driven tuning procedure to select h_3 is provided in the simulation section.

Once the estimated cluster labels $\hat{\ell}$ are obtained, we conduct signal recovery on the sequence $\hat{\mathbf{Y}}_i = \hat{\ell}_i \tilde{\mathbf{X}}_i$ for $i = 1, \dots, n$, where $\hat{\mathbf{Y}}_i = (\hat{Y}_{i1}, \dots, \hat{Y}_{ip})^\top$. Let $\hat{\mathbf{Y}}_0 = n^{-1/2} \sum_{i=1}^n \hat{\mathbf{Y}}_i$, where $\hat{\mathbf{Y}}_0 = (\hat{Y}_{01}, \dots, \hat{Y}_{0p})^\top$. For a given block I , let $\hat{Y}_i(I) = |I|^{-1/2} \sum_{j \in I} \hat{Y}_{ij}$ and $\hat{Y}_0(I) = |I|^{-1/2} \sum_{j \in I} \hat{Y}_{0j}$ be the aggregated feature over I . Let $\hat{\sigma}^2(I)$ be the pooled sample variance of the estimated groups $\{\hat{Y}_i(I) : \hat{\ell}_i = 1\}$ and $\{\hat{Y}_i(I) : \hat{\ell}_i = -1\}$. Recall that $\mathbb{J}_p(h_1) = \{(j, \dots, j+h) : 1 \leq h \leq h_1, 1 \leq j \leq p-h\}$. Take the threshold $t_p = \sqrt{4 \log(ph_1)}$. By the moderate deviation result, if $\delta = \mathbf{0}$, we have $\mathbb{P}(\cup_{I \in \mathbb{J}_p(h_1)} \{|\hat{Y}_0(I)|/\hat{\sigma}(I) > t_p\}) \rightarrow 0$ as $n, p \rightarrow \infty$. Thus, we identify the blocks with nonzero means by $\mathbb{Q}^{(1)} = \{I \in \mathbb{J}_p(h_1) : |\hat{Y}_0(I)|/\hat{\sigma}(I) > t_p\}$. Similar to the CFA procedure for signal identification based on $|W_0(I)|$, we use an iterative step-down procedure based on $|\hat{Y}_0(I)|$ to identify the signal blocks.

Algorithm S3 in the SM outlines the proposed post-clustering signal recovery procedure.

We use $\hat{\mathbb{I}}_{\text{ma}} = \{\hat{I}_1, \hat{I}_2, \dots, \hat{I}_{\hat{m}}\}$ to denote the $\hat{m}$ identified blocks from this procedure.

The following theorem shows that it can consistently identify all true signal blocks.

Theorem 4 *Under Assumptions 1 and 2, Assumption 3 with $c_0 = 1$, $l_{\max} < h_1 < d_0$, $\max\{\|\Sigma\|_2, \|\Sigma^{-1}\|_2\} \leq C$ and $\log(ph_1) = o(n^{1/4})$, given a clustering result $\hat{\ell}$ satisfying $\mathbb{P}(\hat{\ell} = \pm \ell) = 1 - o(1)$, if $\sqrt{n} \min_{1 \leq g \leq m} |\mu_{0,g}| > c\sqrt{\log p}$ for a constant $c > 0$, then we have $\mathbb{P}(\hat{m} = m) \rightarrow 1$ and $\mathbb{P}(\max_{I \in \mathbb{I}_0} \min_{\hat{I} \in \hat{\mathbb{I}}_{\text{ma}}} D(\hat{I}, I) > \zeta) \rightarrow 0$ for any $\zeta > 0$.*

Similar to Theorem 2, we also need equal signal strength in each block for consistent post-clustering signal recovery. The condition $\sqrt{n} \min_g |\mu_{0,g}| > c\sqrt{\log p}$ on signal strength is standard, requiring that the aggregate signal within each block goes to infinity. The proposed method extends the approach in Jeng et al. (2010) for IID Gaussian data with the isotropic covariance matrix to sub-Gaussian data with unknown clusters. Note that the MA-PCA and associated signal recovery procedures for dense block signals work under general covariances, which do not require the bandable covariances assumption in (3.1).

5 Minimax lower bounds

In this section, we establish the statistical and computational minimax lower bounds (SMLB and CMLB) for clustering and signal recovery, and show the proposed CFA-PCA and MA-PCA procedures are computationally minimax optimal. Exact k -means and subset selection are NP-hard, which cannot be solved in polynomial time of n and p . The sparse aggregation method (Jin et al., 2017) applied to $\{\tilde{X}_i(I)\}$ over different blocks is also NP-hard. In contrast, methods based on PCA are computationally feasible. The SMLBs provide the regions of signals where any clustering or signal recovery procedure would fail. While, the computational minimax results are restricted to polynomial time methods.

Let $\lfloor \cdot \rfloor$ denote the floor function. In order to derive the sharp minimax boundaries, we consider the model in (2.1) with $F = N(\mathbf{0}, \mathbf{I}_p)$ for $\mathbf{Z}_i$ and $\varpi \boldsymbol{\mu}_1 + \boldsymbol{\mu}_{-1} = \mathbf{0}$ in this section, indicating the overall mean is removed from the sample. Then, $\boldsymbol{\delta} = (1 + \varpi)\boldsymbol{\mu}_1 = -(1 + \varpi)/\varpi \boldsymbol{\mu}_{-1}$. We consider $\boldsymbol{\delta}$ from the class

$$\begin{aligned} \mathcal{U}(\alpha, \beta, r) = \{ \boldsymbol{\delta} : \text{there exist } m = \lfloor p^{1-\alpha-\beta} \rfloor \text{ blocks with } |I_{0,g}| = \lfloor \rho_g p^\alpha \rfloor, \\ \delta_j = \tau_g \text{ for } j \in I_{0,g} \text{ and } |\tau_g| = \tau = p^{-r}, \delta_j = 0 \text{ otherwise,} \\ \text{and } d_0 > \rho_0 \log(p) p^\alpha \}, \end{aligned} \quad (5.1)$$

where α , β and r are the block size, signal sparsity and signal strength parameters, respectively, satisfying $\alpha \in [0, 1)$, $\beta, r \in (0, 1)$, $0 < \alpha + \beta < 1$ and $\rho_0, \rho_1, \dots, \rho_m > 0$. Here, we impose the specific parameterization $m = \lfloor p^{1-\alpha-\beta} \rfloor$, $|I_{0,g}| \asymp p^\alpha$ and $|\delta_j| = p^{-r}$ for all $j \in S(\boldsymbol{\delta})$ to the general setting in Section 2. We focus on the high-dimensional setting $n = p^\theta$ for $0 < \theta < 1 - \alpha$, namely, $n = o(p^{1-\alpha})$, where $p^{1-\alpha}$ is the dimension of the aggregated variables by blocks of size p^α . Under the class $\mathcal{U}(\alpha, \beta, r)$, $\beta \in ((1 - \alpha)/2, 1 - \alpha)$ and $\beta \in (0, (1 - \alpha)/2)$ correspond to $m \ll \sqrt{p/l_a}$ and $m \gg \sqrt{p/l_a}$, representing the sparse and dense block signal regimes, respectively. The signal structure under $\mathcal{U}(\alpha, \beta, r)$ is similar to the setting of block signals for hypothesis testing in Kou and Walther (2022), and follows the parameterization of signals in Donoho and Jin (2004); Jin et al. (2017).

For the estimated labels $\hat{\boldsymbol{\ell}} = (\hat{\ell}_1, \dots, \hat{\ell}_n)^\top$ from a clustering procedure, we measure its performance using the expected Hamming distance, defined as

$$\text{Hamm}_{\text{clu}}(\hat{\boldsymbol{\ell}}; \boldsymbol{\delta}, \varpi) = \inf_{\pi = \pm 1} n^{-1} \mathbb{E} |\{i : \hat{\ell}_i \neq \pi \ell_i\}|, \quad (5.2)$$

where the expectation is taken with respect to the class labels $\boldsymbol{\ell}$ and the random noise $\mathbf{Z}$ in (2.1). Similarly, for the estimated signal set $\hat{S}$ from a signal recovery procedure, we measure its performance using the normalized expected Hamming distance, defined as

$$\text{Hamm}_{\text{sig}}(\hat{S}; \boldsymbol{\delta}, \varpi) = (p\varepsilon)^{-1} \mathbb{E} |\hat{S} \Delta S(\boldsymbol{\delta})|, \quad (5.3)$$

where $S(\boldsymbol{\delta})$ is the true signal set of $\boldsymbol{\delta}$, $\varepsilon = |S(\boldsymbol{\delta})|/p = s/p$ is the proportion of signals out of p variables, and $A \Delta B = (A \setminus B) \cup (B \setminus A)$ denotes the difference between two sets.

We use the low-degree polynomial conjecture (Hopkins, 2018) to establish our computational minimax lower bounds (CMLBs). This conjecture has been widely applied to various problems (Kunisky et al., 2022; Schramm and Wein, 2022). The CMLBs for clustering and signal recovery can be developed based on the CMLB for the hypotheses testing problem

$$H_0^{(p)} : \mathbf{X}_i \stackrel{\text{iid}}{\sim} N(\mathbf{0}, \mathbf{I}_p) \text{ vs. } H_1^{(p)} : \mathbf{X}_i \stackrel{\text{iid}}{\sim} \frac{\varpi}{1+\varpi} N\left(\frac{\boldsymbol{\delta}}{1+\varpi}, \mathbf{I}_p\right) + \frac{1}{1+\varpi} N\left(-\frac{\varpi \boldsymbol{\delta}}{1+\varpi}, \mathbf{I}_p\right). \quad (5.4)$$

Löffler et al. (2022); Lei et al. (2024) used similar approaches of deriving computational lower bounds through hypotheses testing for clustering under sparse non-block signals and network analysis, respectively. Let $\mathbb{P}_0$ and $\mathbb{P}_1$ be the joint distribution of the data $\{\mathbf{X}_i\}$ under the null and alternative hypotheses in (5.4), respectively. Let $\Lambda(\mathbf{X}) = d\mathbb{P}_1(\mathbf{X})/d\mathbb{P}_0(\mathbf{X})$ be the likelihood ratio, $\Lambda_{(D)}$ be the orthogonal projection of $\Lambda(\mathbf{X})$ onto the linear subspace of polynomials of degree at most D , and $\mathbb{E}_0(\cdot)$ be the expectation under the null distribution $\mathbb{P}_0$. The low-degree polynomial conjecture is stated as follows.

Conjecture 1 (Low-degree polynomial conjecture) *If $\mathbb{E}_0\{\Lambda_{(D)}^2\} = 1+o(1)$, with $D = \log^{1.01}(p)$, then $\mathbb{P}_0$ and $\mathbb{P}_1$ are indistinguishable by any polynomial-time algorithm.*

Let $c_1(\theta, \alpha) = (1 - \theta - \alpha)/2$, $c_2(\theta, \alpha) = 1 - \theta/2 - \alpha$ and $\mathbf{1}(\cdot)$ be the indicator function. For fixed θ, α , introduce $\eta_{\theta, \alpha}^{\text{clu}}(\beta)$ and $\eta_{\theta, \alpha}^{\text{sig}}(\beta)$ for the statistical minimax lower bounds, where

$$\begin{aligned} \eta_{\theta, \alpha}^{\text{clu}}(\beta) &= \{(1 + \theta + \alpha - 2\beta)/4\} \mathbf{1}(\beta < c_1(\theta, \alpha)) \\ &\quad + \{(\theta + \alpha)/2\} \mathbf{1}(c_1(\theta, \alpha) < \beta < 2c_1(\theta, \alpha)) + \{(1 - \beta)/2\} \mathbf{1}(2c_1(\theta, \alpha) < \beta < 1 - \alpha), \\ \eta_{\theta, \alpha}^{\text{sig}}(\beta) &= \{(\theta + \alpha)/2\} \mathbf{1}(\beta < 2c_1(\theta, \alpha)) + \{(1 + \theta + \alpha - \beta)/4\} \mathbf{1}(2c_1(\theta, \alpha) < \beta < 1 - \alpha). \end{aligned}$$

The CMLB $\tilde{\eta}_{\theta, \alpha}^{\text{hyp}}(\beta)$ for testing the hypotheses in (5.4) is derived in the SM using Conjecture 1, where $\tilde{\eta}_{\theta, \alpha}^{\text{hyp}}(\beta)$ is defined in (S.5.4) in the SM. It is shown in Lemma S10 in the SM

that the CMLBs for clustering and signal recovery are no larger than that for testing (5.4). Meanwhile, a CMLB is no larger than the corresponding SMLB. Therefore, we take $\tilde{\eta}_{\theta,\alpha}^{\text{clu}}(\beta) = \min\{\eta_{\theta,\alpha}^{\text{clu}}(\beta), \tilde{\eta}_{\theta,\alpha}^{\text{hyp}}(\beta)\}$ and $\tilde{\eta}_{\theta,\alpha}^{\text{sig}}(\beta) = \min\{\eta_{\theta,\alpha}^{\text{sig}}(\beta), \tilde{\eta}_{\theta,\alpha}^{\text{hyp}}(\beta)\}$ for our computational minimax results, where

$$\begin{aligned}\tilde{\eta}_{\theta,\alpha}^{\text{clu}}(\beta) &= \{(1 + \theta + \alpha - 2\beta)/4\} \mathbf{1}(\beta < (1 - \alpha)/2) \\ &\quad + (\theta/4 + \alpha/2) \mathbf{1}((1 - \alpha)/2 < \beta < c_2(\theta, \alpha)) + \{(1 - \beta)/2\} \mathbf{1}(c_2(\theta, \alpha) < \beta < 1 - \alpha), \\ \tilde{\eta}_{\theta,\alpha}^{\text{sig}}(\beta) &= \{(\theta + \alpha)/2\} \mathbf{1}(\beta < c_1(\theta, \alpha)) + \{(1 + \theta + \alpha - 2\beta)/4\} \mathbf{1}(c_1(\theta, \alpha) < \beta < (1 - \alpha)/2) \\ &\quad + (\theta/4 + \alpha/2) \mathbf{1}((1 - \alpha)/2 < \beta < 1 - \alpha).\end{aligned}$$

The following two theorems state the statistical and computational minimax lower bounds for clustering and signal recovery under the block signal setting.

Theorem 5 *Under the model in (2.1) with $\mathbf{Z}_i \sim N(\mathbf{0}, \mathbf{I}_p)$, given $\alpha \in [0, 1)$, $\theta \in (0, 1 - \alpha)$, $\beta \in (0, 1 - \alpha)$ and $\varpi > 0$, we have $\lim_{p \rightarrow \infty} \sup_{\boldsymbol{\delta} \in \mathcal{U}(\alpha, \beta, r)} \text{Hamm}_{\text{clu}}(\hat{\boldsymbol{\ell}}; \boldsymbol{\delta}, \varpi) \geq \min\{1, \varpi\}/(1 + \varpi)$ for (1) any clustering procedure $\hat{\boldsymbol{\ell}}$ if $r > \eta_{\theta,\alpha}^{\text{clu}}(\beta)$ and (2) any polynomial-time clustering procedure $\hat{\boldsymbol{\ell}}$ if $r > \tilde{\eta}_{\theta,\alpha}^{\text{clu}}(\beta)$ under Conjecture 1.*

Theorem 6 *Under the model in (2.1) with $\mathbf{Z}_i \sim N(\mathbf{0}, \mathbf{I}_p)$, given $\alpha \in [0, 1)$, $\theta \in (0, 1 - \alpha)$, $\beta \in (0, 1 - \alpha)$ and $\varpi > 0$, we have $\lim_{p \rightarrow \infty} \sup_{\boldsymbol{\delta} \in \mathcal{U}(\alpha, \beta, r)} \text{Hamm}_{\text{sig}}(\hat{S}; \boldsymbol{\delta}, \varpi) \geq C > 0$ for (1) any signal recovery procedure $\hat{S}$ if $r > \eta_{\theta,\alpha}^{\text{sig}}(\beta)$ and (2) any polynomial-time signal recovery procedure $\hat{S}$ if $r > \tilde{\eta}_{\theta,\alpha}^{\text{sig}}(\beta)$ under Conjecture 1.*

Theorems 5 and 6 characterize both the statistical and computational minimax lower bounds for clustering and signal recovery under block signals. They demonstrate that when $r > \eta_{\theta,\alpha}^{\text{clu}}(\beta)$ and $r > \eta_{\theta,\alpha}^{\text{sig}}(\beta)$, no statistical algorithm can achieve consistent clustering and signal recovery, respectively. Note that the signal strength τ decreases with the increase of r in the class $\mathcal{U}(\alpha, \beta, r)$, and the error rate $\min\{1, \varpi\}/(1 + \varpi)$ can be achieved by simply

guessing $\hat{\ell}_i = 1$ or $\hat{\ell}_i = -1$ for all i . In particular, when $\alpha = 0$, representing signals without block structures, the derived SMLBs align with those in Jin et al. (2017) under a Bayesian setting of randomly located signals. The two theorems also show that $\tilde{\eta}_{\theta,\alpha}^{\text{clu}}(\beta)$ and $\tilde{\eta}_{\theta,\alpha}^{\text{sig}}(\beta)$ are the CMLBs for clustering and signal recovery. Namely, no polynomial-time algorithm can achieve consistent clustering and signal recovery when $r > \tilde{\eta}_{\theta,\alpha}^{\text{clu}}(\beta)$ and $r > \tilde{\eta}_{\theta,\alpha}^{\text{sig}}(\beta)$, respectively. This delineates a “region of impossibility” and provides a lower bound on the necessary signal strength for successful clustering and signal recovery under block signals and polynomial-time computational constraint. Meanwhile, note that the four functions $\eta_{\theta,\alpha}^{\text{clu}}(\beta)$, $\eta_{\theta,\alpha}^{\text{sig}}(\beta)$, $\tilde{\eta}_{\theta,\alpha}^{\text{clu}}(\beta)$ and $\tilde{\eta}_{\theta,\alpha}^{\text{sig}}(\beta)$ increase as θ or α increases, which means that the impossible regions shrink with increasing sample size and length of signal blocks.

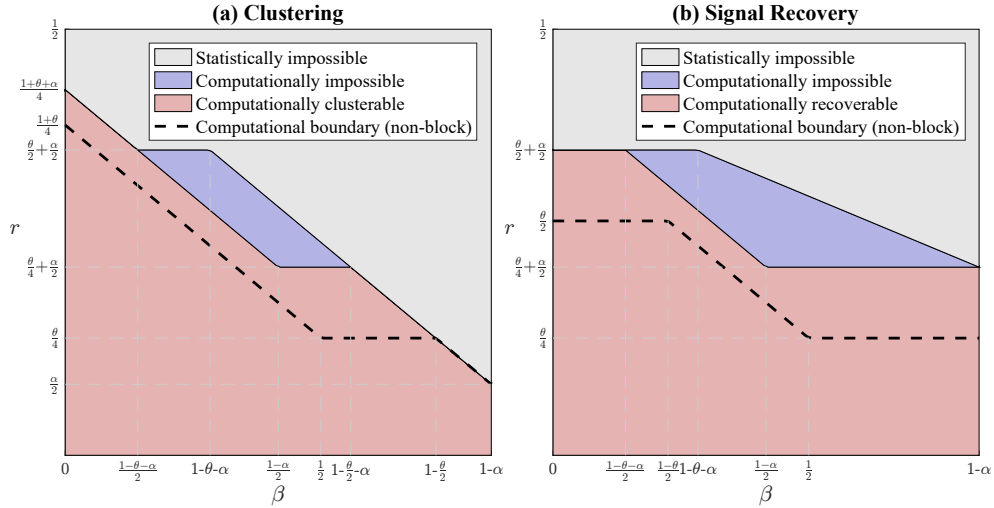


Figure 1: Phase transition of clustering in panel (a) and signal recovery in panel (b). Gray region: consistent clustering or signal recovery is impossible by any algorithm. Purple region: consistent clustering or signal recovery is impossible under the polynomial computation time constraint. Red region: consistent clustering or signal recovery can be achieved by a polynomial-time algorithm. Black dash line: CMLBs under non-block signals ($\alpha = 0$).

Panels (a) and (b) in Figure 1 illustrate the SMLBs and CMLBs. From those plots, there

is a statistical-computational gap that $\tilde{\eta}_{\theta,\alpha}^{\text{clu}} < \eta_{\theta,\alpha}^{\text{clu}}$ for clustering in the moderately sparse regime of $(1 - \theta - \alpha)/2 < \beta < 1 - \theta/2 - \alpha$. For signal recovery, a statistical-computational gap exists except the regime of $\beta < (1 - \theta - \alpha)/2$. Figure 1 also shows the computationally impossible regions are much smaller under block signals, which demonstrates the benefit in the minimax lower bounds gained by considering the block structure of signals. Phase diagrams for some specific values of θ, α are shown in Figures S.7 and S.8 of the SM.

It is worth noting that Löffler et al. (2022) derived the computational minimax threshold for the signal-to-noise ratio in the high-dimensional clustering problem, which corresponds to the special case of $\alpha = 0$ in our setting for non-block signals. However, they didn't establish the phase transition diagram, nor study the minimax results for signal recovery. Therefore, their results don't show the statistical-computational gaps in the phase transition diagram, and cannot compare the difficulty of clustering and signal recovery across different sparsity regimes from a minimax view.

Let $\hat{S}_{\text{cfa}} = \cup_{\hat{I} \in \hat{\mathcal{I}}_{\text{cfa}}} \hat{I}$ and $\hat{S}_{\text{ma}} = \cup_{\hat{I} \in \hat{\mathcal{I}}_{\text{ma}}} \hat{I}$ be the sets of identified signals by the two proposed methods. From the results of Theorems 1–4, the following corollary demonstrates the optimality of the proposed CFA-PCA procedure ($\hat{\ell}_{\text{cfa}}$ and $\hat{S}_{\text{cfa}}$) and the MA-PCA procedure ($\hat{\ell}_{\text{ma}}$ and $\hat{S}_{\text{ma}}$) under the Gaussian distribution with identity covariance.

Corollary 1 *Under the model in (2.1) with $\mathbf{Z}_i \sim N(\mathbf{0}, \mathbf{I}_p)$, given $\alpha \in [0, 1)$, $\theta \in (0, 1 - \alpha)$, $\beta \in (0, 1 - \alpha)$ and $\varpi > 0$, choosing $h_1, h_3 \asymp p^\alpha$ and $h_2 = cd_0$ for any $c \in [0, 1)$, then*

$$\lim_{n, p \rightarrow \infty} \sup_{\boldsymbol{\delta} \in \mathcal{U}(\alpha, \beta, r)} \text{Hamm}_{\text{clu}}(\hat{\ell}; \boldsymbol{\delta}, \varpi) \rightarrow 0 \text{ for } r < \tilde{\eta}_{\theta, \alpha}^{\text{clu}}(\beta) \text{ and}$$

$$\lim_{n, p \rightarrow \infty} \sup_{\boldsymbol{\delta} \in \mathcal{U}(\alpha, \beta, r)} \text{Hamm}_{\text{sig}}(\hat{S}; \boldsymbol{\delta}, \varpi) \rightarrow 0 \text{ for } r < \tilde{\eta}_{\theta, \alpha}^{\text{sig}}(\beta),$$

where $\hat{\ell} = \hat{\ell}_{\text{ma}}$, $\hat{S} = \hat{S}_{\text{ma}}$ if $\beta < (1 - \alpha)/2$, and $\hat{\ell} = \hat{\ell}_{\text{cfa}}$, $\hat{S} = \hat{S}_{\text{cfa}}$ if $\beta > (1 - \alpha)/2$.

Note that $\beta > (1 - \alpha)/2$ and $\beta < (1 - \alpha)/2$ correspond to $m \ll \sqrt{p/l_a}$ and $m \gg \sqrt{p/l_a}$ under the class $\mathcal{U}(\alpha, \beta, r)$. Corollary 1 shows that the proposed CFA-PCA and MA-PCA

procedures are computationally minimax optimal for clustering and signal recovery in the sparse and dense block signal regimes, respectively. It demonstrates the CMLBs derived in Theorems 5 and 6 are tight. Table S.8 in the SM summarizes those results.

6 Extension to multi-group clustering

This section extends our methods to K -group clustering for $K > 2$. We consider the same mixture model $\mathbf{X}_i = \boldsymbol{\mu}_{\ell_i} + \mathbf{Z}_i$ in (2.1), where ℓ_i follows a categorical distribution

$$\ell_i \sim \text{Categorical}(\{1, 2, \dots, K\}, \{\pi_1, \pi_2, \dots, \pi_K\}) \quad (6.1)$$

for $\pi_k \in (0, 1)$ and $\sum_{k=1}^K \pi_k = 1$, ℓ_i and $\mathbf{Z}_i$ are independent, and $\boldsymbol{\mu}_k = \mathbb{E}(\mathbf{X}_i \mid \ell_i = k)$. Let $\boldsymbol{\mu} = \sum_{k=1}^K \pi_k \boldsymbol{\mu}_k$ be the overall mean of $\mathbf{X}_i$ and $\tilde{\boldsymbol{\mu}}_k = \boldsymbol{\mu}_k - \boldsymbol{\mu}$ be the centered mean. Let $\mathbf{M} = (\boldsymbol{\mu}_1, \dots, \boldsymbol{\mu}_K)$ and $\mathbf{L}_i = (\mathbf{1}(\ell_i = 1), \dots, \mathbf{1}(\ell_i = K))$. Then, $\mathbf{X}_i = \mathbf{L}_i \mathbf{M} + \mathbf{Z}_i$ and

$$\text{Cov}(\mathbf{X}_i) = \text{Cov}(\mathbf{L}_i \mathbf{M}) + \text{Cov}(\mathbf{Z}_i) = \sum_{k=1}^K \pi_k \tilde{\boldsymbol{\mu}}_k \tilde{\boldsymbol{\mu}}_k^T + \boldsymbol{\Sigma}, \quad (6.2)$$

where $\sum_{k=1}^K \pi_k \tilde{\boldsymbol{\mu}}_k \tilde{\boldsymbol{\mu}}_k^T$ is the signal part for multi-group clustering, which is proportional to $\boldsymbol{\delta} \boldsymbol{\delta}^T$ for two-group clustering, and $\boldsymbol{\Sigma}$ is the noise part. As CFA-PCA essentially uses covariances for variable selection, the same procedure can be applied to multi-group problems.

Similar to the setting of two-group clustering, we consider a block structure for $\tilde{\boldsymbol{\mu}}_k$. Assume the nonzero components of $\tilde{\boldsymbol{\mu}}_k$ are distributed in non-overlapping blocks $\mathbb{I}_0 = \{I_{0,g} : 1 \leq g \leq m\}$, where $l_g = |I_{0,g}|$ and $\tilde{\mu}_{kj} = 0$ for $k = 1, \dots, K$ if j is not in those blocks. Let d_0 be the minimum distance among the blocks in $\mathbb{I}_0$.

Assumption 4 For each $I_{0,g} \in \mathbb{I}_0$, $k, k' = 1, \dots, K$ and $k \neq k'$, we have $c_0^{-1} \delta_{k,g} \leq |\tilde{\mu}_{kj}| \leq c_0 \delta_{k,g}$ for $\delta_{k,g} \geq 0$ and $c_0 \geq 1$, and $\text{sign}(\tilde{\mu}_{kj} - \tilde{\mu}_{k'j}) = e_{kk',g}$ for $j \in I_{0,g}$ and $e_{kk',g} \in \{-1, 0, 1\}$.

For a given block I , let $\tilde{\mu}_k(I) = |I|^{-1/2} \sum_{j \in I} \tilde{\mu}_{kj}$ be the accumulated signal strength of the block I and group k . Let $\tilde{\delta}(I_1, I_2) = \sum_{k=1}^K \pi_k \tilde{\mu}_k(I_1) \tilde{\mu}_k(I_2)$ and $\tilde{\delta}(I) = \tilde{\delta}(I, I) =$

$\sum_{k=1}^K \pi_k \tilde{\mu}_k^2(I)$. We consider the non-trivial case that $\{\tilde{\delta}(I_{0,g})\}_{g=1}^m$ are bounded. Otherwise, χ^2 statistics can easily identify the signal blocks. Due to the expression in (6.2), the same CFA procedure introduced in Section 3 can identify nonzero $\tilde{\delta}(I_1, I_2)$ by $\{W_i(I_1, I_2)\}_{i=1}^n$. Using the same notation as two-group clustering, let $\hat{\mathbb{I}}_{\text{cfa}} = \{\hat{I}_1, \dots, \hat{I}_{\hat{m}}\}$ denote the identified blocks by the CFA method, and $\hat{\mathbf{Y}}^{(\text{cfa})} = (\hat{\mathbf{Y}}_{(1)}^{(\text{cfa})}, \dots, \hat{\mathbf{Y}}_{(\hat{m})}^{(\text{cfa})})$ be the aggregated data matrix of the selected features. Let $\hat{\mathbf{U}}^{(\text{cfa})} \in \mathbb{R}^{n \times (K-1)}$ denote the top $(K-1)$ eigenvectors of $\hat{\mathbf{Y}}^{(\text{cfa})} \hat{\mathbf{Y}}^{(\text{cfa})\top}$. The multi-group CFA-PCA method applies the $(1+\varepsilon)$ -approximate k -means algorithm (Kumar et al., 2004) to the rows of $\hat{\mathbf{U}}^{(\text{cfa})}$, which produces the estimated cluster labels $\hat{\ell}_{\text{cfa}}$. For MA-PCA, let $\mathbf{Y}^{(\text{ma})}$ denote the same column-wise moving average matrix of $\tilde{\mathbf{X}}$ in Section 4. Similarly, the multi-group MA-PCA procedure is the same as that for two-group clustering, except the $(1+\varepsilon)$ -approximate k -means algorithm is applied to the top $(K-1)$ eigenvectors of $\mathbf{Y}^{(\text{ma})} \mathbf{Y}^{(\text{ma})\top}$. Let $\hat{\ell}_{\text{ma}}$ be the estimated labels. The detailed algorithms of the two procedures are provided in Section S7 of the SM.

Let $\text{Hamm}_{\text{clu}}(\hat{\ell}; \{\boldsymbol{\mu}_k\}_{k=1}^K, \{\pi_k\}_{k=1}^K) = \inf_{\zeta \in \mathcal{P}_K} n^{-1} \mathbb{E} |\{i : \hat{\ell}_i \neq \zeta(\ell_i)\}|$ be the clustering error rate of $\hat{\ell}$, where $\zeta(\cdot)$ is a permutation mapping and $\mathcal{P}_K$ denotes the set of all permutations of $\{1, \dots, K\}$. Let $\lambda_1 \geq \dots \geq \lambda_K$ be the leading K eigenvalues of the matrix $\mathbf{M}^\top \mathbf{M}$. The following two theorems establish the consistency of the multi-group CFA-PCA and MA-PCA procedures for clustering.

Theorem 7 *Under Assumptions 1, 2 and 4, $\boldsymbol{\Sigma} \in \mathfrak{U}(\gamma, C)$ and $\max\{\|\boldsymbol{\Sigma}\|_2, \|\boldsymbol{\Sigma}^{-1}\|_2\} \leq C$, if $\sqrt{n} l_{\max} d_0^{-\gamma-1} = o(1)$, $h_1 = c_1 l_{\max}$ and $h_2 = c_2 d_0$ for any constants $c_1 \geq 1$ and $c_2 \in (0, 1)$, $\log(pl_{\max}) = o(n^{1/4})$, $\lambda_1/\lambda_K < c'$ for a constant $c' > 0$, and*

$$\min_{1 \leq g \leq m} \max_{g' \neq g} |\tilde{\delta}(I_{0,g}, I_{0,g'})| > \frac{c_3 \log p}{\sqrt{n}} \text{ and } \min \left\{ \frac{\sqrt{n}}{\sqrt{m}}, 1 \right\} \min_{k_1 \neq k_2} \|\boldsymbol{\mu}_{k_1} - \boldsymbol{\mu}_{k_2}\|_2^2 > c_3 \log n \quad (6.3)$$

hold for some $c_3 > 0$, then $\text{Hamm}_{\text{clu}}(\hat{\ell}_{\text{cfa}}, \{\boldsymbol{\mu}_k\}, \{\pi_k\}) \rightarrow 0$.

Theorem 8 Under Assumptions 1-2 and 4, if $\lambda_1/\lambda_K < c'$ for a constant $c' > 0$ and

$$\min \left\{ \frac{\sqrt{n}}{\sqrt{ph_3}}, \frac{1}{h_3} \right\} \|\Sigma\|_2^{-1} \min_{k_1 \neq k_2} \left\{ \sum_{g=1}^m \min\{h_3, l_g\} (\tilde{\mu}_{k_1}(I_{0,g}) - \tilde{\mu}_{k_2}(I_{0,g}))^2 \right\} > c_4 \log n, \quad (6.4)$$

for a constant $c_4 > 0$, then $\text{Hamm}_{\text{clu}}(\hat{\ell}_{\text{ma}}; \{\boldsymbol{\mu}_k\}, \{\pi_k\}) \rightarrow 0$.

The signal-to-noise ratio conditions in (6.3) and (6.4) align with those for two-group clustering in Theorems 1 and 3. For multi-group clustering, we require the means of any pair of groups are well-separated so that they can be consistently clustered. Particularly, for the multi-group CFA-PCA procedure, $\tilde{\delta}(I_1, I_2)$ is the block accumulation of $\sum_{k=1}^K \pi_k \tilde{\boldsymbol{\mu}}_k \tilde{\boldsymbol{\mu}}_k^T$. The first condition in (6.3) imposes a minimum signal strength requirement on $\max_{g' \neq g} |\tilde{\delta}(I_{0,g}, I_{0,g'})|$ for every $I_{0,g} \in \mathbb{I}_0$, ensuring $I_{0,g}$ can be identified using the cross-block covariance with at least one other signal block.

7 Extension to tensor data

This section extends the proposed methods to matrix- and tensor-valued data, commonly observed in geoscience and medical imaging studies. We only consider two-group clustering for this section. Suppose we observe n independent order- q tensors with dimensions $p_1, \dots, p_q$, denoted by $\{\mathcal{X}_i = (X_{i,j_1 \dots j_q})\}_{i=1}^n$, where j_t denotes the j_t th variable observed in the t th mode of the tensor for $1 \leq j_t \leq p_t$. Let $p = \prod_{t=1}^q p_t$ be the total dimension of the tensor. Extending (2.1) to tensor data, we assume $\{\mathcal{X}_i\}$ satisfy the cluster model $\mathcal{X}_i = \mathcal{U}_{\ell_i} + \mathcal{Z}_i$, for $i = 1, 2, \dots, n$ and $\ell_i \in \{-1, 1\}$, where $\mathcal{U}_{\pm 1} = \mathbb{E}(\mathbf{X}_i | \ell_i = \pm 1)$ are the signals for clustering, and $\mathcal{Z}_i$ represents random noise. Our primary goal is to estimate the cluster labels $\boldsymbol{\ell} = (\ell_1, \dots, \ell_n)^T$ and recover the signal set $\mathcal{S}(\mathcal{U}) = \{(j_1, j_2, \dots, j_q) : \mathcal{U}_{j_1, j_2, \dots, j_q} \neq 0\}$, where $\mathcal{U} = \mathcal{U}_1 - \mathcal{U}_{-1}$. Let $\mathcal{I}_{0,1}, \dots, \mathcal{I}_{0,m}$ denote the m non-overlapping tensor blocks of nonzero mean difference, where $\mathcal{I}_{0,g} = I_{0,g}^{(1)} \times \dots \times I_{0,g}^{(q)}$ and $I_{0,g}^{(t)} = \{j_g^{(t)}, \dots, j_g^{(t)} + l_g^{(t)} - 1\}$ is a block in

the t th mode for $t = 1, \dots, q$ and $g = 1, \dots, m$. Theorem S2 in SM shows that the SMLBs and CMLBs derived for vector-valued data can be extended to tensor-valued data.

In the following, we extend the proposed MA-PCA and CFA-PCA procedures to tensor-valued data. Let $\tilde{X}_{i,j_1\dots,j_q} = X_{i,j_1\dots,j_q} - \sum_{i=1}^n X_{i,j_1\dots,j_q}/n$ be the centered $\mathcal{X}_i$, and $\tilde{\mathbf{X}}_{(j_1\dots,j_q)} = (\tilde{X}_{1,j_1\dots,j_q}, \dots, \tilde{X}_{n,j_1\dots,j_q})^T$. For the CFA-PCA method, let $\mathcal{J}_p(h_1) = \mathbb{J}_{p_1}(h_1) \times \dots \times \mathbb{J}_{p_q}(h_1)$, where $\mathbb{J}_{p_t}(h_1) = \{\{j, \dots, j+h\} : 1 \leq h \leq h_1, 1 \leq j < j+h \leq p_t\}$. For any pair of blocks $\mathcal{I}_1, \mathcal{I}_2 \in \mathcal{J}_p(h_1)$, let $\tilde{X}_i(\mathcal{I}_1) = \sum_{(j_1, \dots, j_q) \in \mathcal{I}_1} \tilde{X}_{i,j_1\dots,j_q} / \sqrt{|\mathcal{I}_1|}$ and $W_i(\mathcal{I}_1, \mathcal{I}_2) = \tilde{X}_i(\mathcal{I}_1) \tilde{X}_i(\mathcal{I}_2)$. Similar to (3.2), for each $\mathcal{I}_1 \in \mathcal{J}_p(h_1)$, we can define the maximal cross-block product $W_0(\mathcal{I}_1)$ based on $W_i(\mathcal{I}_1, \mathcal{I}_2)$. Let $\mathcal{E}(\mathcal{I}, \tilde{c}) = \mathcal{E}_{p_1}(I^{(1)}, \tilde{c}) \times \dots \times \mathcal{E}_{p_q}(I^{(q)}, \tilde{c})$ be the expanding operator for tensor blocks, where $\tilde{c}$ is the expanded window length and $\mathcal{E}_p(I^{(t)}, \tilde{c})$ is the expanded block of $I^{(t)}$. Using $W_0(\mathcal{I})$, the CFA method in Section 3 can be extended to tensor-valued data to identify the tensor blocks with nonzero means. Once the estimated blocks $\hat{\mathcal{I}}_1, \dots, \hat{\mathcal{I}}_{\hat{m}}$ are obtained from the CFA procedure, we construct the aggregated data matrix for those selected blocks as $\hat{\mathbf{Y}}^{(\text{cfa})} = (\hat{\mathbf{Y}}_{(1)}^{(\text{cfa})}, \dots, \hat{\mathbf{Y}}_{(\hat{m})}^{(\text{cfa})})$, where $\hat{\mathbf{Y}}_{(k)}^{(\text{cfa})} = |\hat{\mathcal{I}}_k|^{-1/2} \sum_{(j_1, \dots, j_q) \in \hat{\mathcal{I}}_k} \tilde{\mathbf{X}}_{(j_1\dots,j_q)}$ for $k = 1, \dots, \hat{m}$. The cluster labels ℓ can be estimated by $\hat{\ell}_{\text{cfa}} = \text{sign}(\hat{\xi}_{\text{cfa}})$, where $\hat{\xi}_{\text{cfa}}$ is the first eigenvector of the matrix $\hat{\mathbf{Y}}^{(\text{cfa})} \hat{\mathbf{Y}}^{(\text{cfa})T}$.

For the tensor-valued MA-PCA method, let $\mathbf{Y}_{(g_1\dots,g_q)}^{(\text{ma})} = \frac{1}{\sqrt{h_3^q}} \sum_{j_1=g_1}^{g_1+h_3-1} \dots \sum_{j_q=g_q}^{g_q+h_3-1} \tilde{\mathbf{X}}_{(j_1\dots,j_q)}$ be the moving average with a window length h_3 on each mode. Gather $\mathbf{Y}_{(g_1\dots,g_q)}^{(\text{ma})}$ into a large data matrix $\mathbf{Y}^{(\text{ma})} \in \mathbb{R}^{n \times \tilde{p}}$, where $\tilde{p} = \prod_{t=1}^q (p_t - h_3 + 1)$. The MA-PCA estimate of ℓ is given by $\hat{\ell}_{\text{ma}} = \text{sign}(\hat{\xi}_{\text{ma}})$, where $\hat{\xi}_{\text{ma}}$ is the first eigenvector of the matrix $\mathbf{Y}^{(\text{ma})} \mathbf{Y}^{(\text{ma})T}$.

We conjecture the tensor MA-PCA and CFA-PCA procedures can also achieve the CMLBs in Theorem S2 for clustering and signal recovery under the dense and sparse signal regimes, respectively. We leave the theoretical justification in future work. In Section 8, we conduct simulation experiments using matrix-valued data, demonstrating good perfor-

manances of the proposed procedures under different regimes of signals.

8 Simulation

In this section, we evaluate the proposed methods MA-PCA and CFA-PCA for clustering and signal recovery, and compare them with IF-PCA (Jin and Wang, 2016), spectral clustering and k -means numerically. Note that the latter three algorithms are not designed for block signals. We combine them with the post-clustering block identification procedure of MA-PCA in Algorithm S3 to evaluate their performance on signal recovery in our setting.

We considered two-group clustering for vector-valued data in this section. Results for matrix-valued data are reported in the SM. We generated $\mathbf{X}_i = \boldsymbol{\mu}_{\ell_i} + \mathbf{Z}_i$ with $\boldsymbol{\mu}_1 = -\boldsymbol{\mu}_{-1}$ and $\ell_i \sim 2\text{Bernoulli}(1/2) - 1$ independently for $i = 1, \dots, n$, where $\boldsymbol{\mu}_{\pm 1} \in \mathbb{R}^p$ with m signal blocks and other entries being 0, and $\mathbf{Z}_i = (Z_{i1}, \dots, Z_{ip})^\top$ with $Z_{ij} \stackrel{\text{iid}}{\sim} N(0, 1)$. We set $n = 2\lceil p^\theta/2 \rceil$ for $\theta = 0.4$ and $m = \lfloor p^{1-\alpha-\beta} \rfloor$, where $\lfloor \cdot \rfloor$ and $\lceil \cdot \rceil$ denote the floor and ceil functions, respectively. For the cases of dense and sparse block signals, we set $\alpha = 0.5, \beta = 0.24$ and $\alpha = 0.3, \beta = 0.6$, respectively. The block size ranged from $l_{\min} = \lfloor \rho_{\min} p_1^\alpha \rfloor$ to $l_{\max} = \lceil \rho_{\max} p^\alpha \rceil$ with a minimum gap between two blocks $d_0 = \lceil \rho_0 p^\alpha \rceil$, where $\rho_{\min} = 0.8$, $\rho_{\max} = 1.25$ and $\rho_0 = 1.5$. The locations of the signal blocks were randomly selected in $\boldsymbol{\mu}_1$. We first randomly chose the center of the first block from the set $\{1, \dots, p\}$ excluding the border band of size $l_{\max}/2$. Then, a block size was randomly drawn from a uniform distribution over $[l_{\min}, l_{\max}]$ to generate $I_{0,1}$. We removed the region $\mathcal{E}_p(I_{0,1}, d_0)$ from the candidate set of grids for selecting the next block. The rest of the blocks were generated in the same manner sequentially, until a total of m blocks were identified. For the entries in the selected blocks, we assigned $\mu_{1,j} = \tau_g$ for $j \in I_{0,g}$, where $\tau_g = \tau$ or $-\tau$ with equal probability. We set the dimensions $p = 1000, 2500, 5000$ and $n = 22, 38, 68$. For

each combination of (α, β) and (n, p) , we considered a sequence of values for the signal strength τ . The specific settings of $\boldsymbol{\mu}_{\pm 1}$ are provided in Table S.1 in the SM. Once $\boldsymbol{\mu}_{\pm 1}$ was generated, it was kept fixed through 500 repetitions for each setting.

We propose a data-driven method to select the window sizes. For MA-PCA, let $\hat{s}(h_1, h_3)$ be the number of identified signals using the hyperparameters h_1 and h_3 . From the results of sensitivity analysis in Tables S.2–S.4 in the SM, we find that the smallest signal recovery loss typically corresponded to the smallest clustering loss. The post-clustering signal recovery is based on $\hat{\mathbf{Y}}_0 = n^{-1/2} \sum_{i=1}^n \hat{\ell}_i \tilde{\mathbf{X}}_i$. If the clustering is inaccurate, observations from different clusters are combined with incorrect signs, which causes cancellation and weakens the block signals; consequently, fewer true signals can be recovered. Meanwhile, if the clustering results are exactly correct such that $\hat{\ell}_i = \ell_i$ for all i , using moderate deviation results, the chosen threshold $\sqrt{4 \log(ph_1)}$ for $|\hat{Y}_0(I)|/\hat{\sigma}(I)$ guarantees no false positives to be identified from $\hat{Y}_0(I)$ if $\delta(I) = 0$. This motivates us to use the number of significant variables $\hat{s}(h_1, h_3)$ as the criterion. Given a candidate set $\mathcal{H} = \{(h_1, h_3) : 1 \leq h_1 \leq h_3 \leq h_{\max}\}$, let $\hat{s}_{\max} = \max_{(h_1, h_3) \in \mathcal{H}} \hat{s}(h_1, h_3)$. We select the window size $(\hat{h}_1, \hat{h}_3)$ as $(\hat{h}_1, \hat{h}_3) = \operatorname{argmin}_{(h_1, h_3) \in \mathcal{H}} \{h_1 + h_3 : \hat{s}(h_1, h_3) > (1 - \epsilon)\hat{s}_{\max}\}$, where ϵ is a small positive constant. In the simulation, we set $\epsilon = 0.01$ and $h_{\max} = d_0$. A sensitivity analysis on $h_{\max}$ was reported in Table S.5 in the SM, along with the results under non-Gaussian distributions. Similarly, although the CFA signal recovery procedure is conducted before clustering, we also use the post-clustering signal recovery result to select the hyperparameters h_1 and h_2 for CFA-PCA, which is described in detail in Section S1.3 of the SM.

For a fair comparison with the existing methods, IF-PCA, spectral clustering and k -means, which do not consider block signals, we also evaluated the proposed methods under non-block signals where the signal components were randomly selected. Under this setting,

$\alpha = 0$ and we set $\beta = 0.3$ and 0.8 for the cases of dense and sparse signals, respectively. The results are reported in Figures S.1 and S.2 in the SM for vector and matrix data, respectively. From those plots, we observe that the clustering errors of MA-PCA and spectral clustering were quite similar and better than CFA-PCA and IF-PCA under the dense signal setting, whereas, the clustering errors of CFA-PCA and IF-PCA were similar and better than MA-PCA and spectral clustering under the sparse signal setting. The performance of k -means was comparable with CFA-PCA and IF-PCA for dense signals, but was the worst for sparse signals. Meanwhile, for signal recovery, CFA-PCA and IF-PCA were comparable and much better than the other three methods under the sparse signal setting. Those results demonstrate that the proposed methods do not suffer an efficiency loss from considering block structures for non-block signals. Their performances were comparable to the existing methods under the non-block setting.

Figure 2 and Figure S.3 in the SM present the results under the block signal settings for vector and matrix data, respectively. The detailed settings for the matrix-valued data are described in Section S1.5 in the SM. From those figures, the proposed CFA-PCA and MA-PCA methods far outperformed the three existing methods without considering the block structures. MA-PCA performed notably better in the dense regime, while CFA-PCA excelled in the sparse regime. This trend became more pronounced as the dimension increased, consistent with our theoretical findings. Those results provide the evidence that the proposed MA-PCA and CFA-PCA methods are superior over the existing methods for clustering and signal recovery under block signals, and the two methods are respectively better suited for dense and sparse block signals.

We also conducted sensitivity analyses for the hyperparameters and error distributions in MA-PCA and CFA-PCA. We report the clustering error, signal recovery error, and com-

putation time under different distributions and hyperparameters, h_1, h_2, h_3 and $h_{\max}$, in Tables S.2–S.5 of the SM. The results indicate that both procedures were stable over a wide range of tuning parameter values, and remained robust under different noise distributions. The empirical running times of MA-PCA and CFA-PCA increased linearly and quadratically with the increase of the window size, respectively, which aligns with the rates of their computation costs with respect to h_1 and h_3 , introduced after (3.3) and (4.1).

9 Case study

We study the yearly change of global temperatures to investigate global warming patterns. We extracted 1949–2023 global annual averages of surface temperature anomalies from the HadCRUT5 dataset (Morice et al., 2021), with $5^\circ \times 5^\circ$ spatial resolution. We applied the proposed methods to matrix-valued observations of yearly temperature differences with dimension 72×36 , where each matrix component corresponds to a spatial grid, and signal blocks are formed by spatially contiguous geographical regions. Algorithms S1 and S2 in the SM provide the detailed implementation of the proposed methods for matrix data.

We applied the CFA-PCA, MA-PCA, IF-PCA, spectral clustering and k -means methods to the data for signal recovery and clustering, categorizing the years into two clusters. Figure 3 reports the results of MA-PCA and CFA-PCA. The results of the other three methods are reported in Figure S.4 in the SM. Of the 74 samples of yearly temperature changes, CFA-PCA assigned 39 to one cluster, and 35 to the other cluster. From the signal recovery map shown in Figure 3, CFA-PCA mainly identified four blocks: northern North America, the Siberia region, the equatorial part of the eastern Pacific, and the eastern hemisphere portion of Antarctica. The first three regions exhibited significant warming signals in the first cluster and cooling signals in the second cluster, while the last block showed

the opposite pattern. Specifically, the equatorial Pacific sector 5°N – 5°S , 120° – 170°W (the Niño 3.4 region) exhibited only very weak local signals that were nonetheless successfully detected by CFA-PCA, while MA-PCA did not identify this region. This is particularly noteworthy because Niño 3.4 is known as a key region for the El Niño Southern Oscillation (ENSO), which is widely regarded as an important driver of global climate variability and long-term warming patterns. The successful identification of the Niño 3.4 region by the CFA-PCA method indicates that it produced a more scientifically reasonable clustering result than the MA-PCA method.

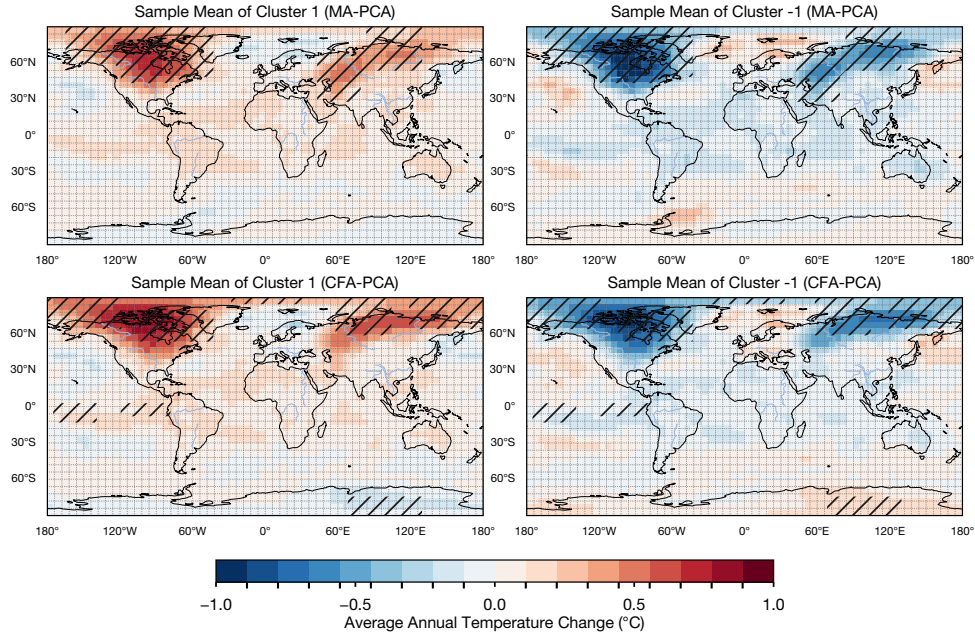


Figure 3: Estimated cluster means of yearly surface temperature changes by MA-PCA (upper panels) and CFA-PCA (bottom panels). Shaded areas indicate identified blocks.

From Figure S.4 in the SM, the two clusters identified by the other three methods, IF-PCA, spectral clustering and k -means, appeared to be dominated by the strong signals over northern Siberia. The means of the two clusters exhibited an opposite pattern in the northeastern Canada and Greenland regions versus northern Siberia. However, these

two regions, collectively known as boreal forests, share very similar climate patterns. This suggests that the results of those methods might lack scientific explanations.

We also calculated the standardized difference of the Niño 3.4 index for the two clusters by different methods. Niño 3.4 index is commonly used to distinguish different types of ENSO. The t -statistics were 2.17, 2.34, 0.05, 0.30 and 1.01 for CFA-PCA, MA-PCA, IF-PCA, spectral clustering and k -means, respectively. Only CFA-PCA and MA-PCA produced clusters that showed a significant difference in the Niño 3.4 index. Those findings suggest that modeling block signals can be essential for uncovering scientifically interpretable signals of global warming and ENSO variability in large-scale earth science data.

10 Discussion

In this paper, we develop both minimax lower bounds and computationally tractable algorithms for high-dimensional clustering under block signals. The two proposed procedures, MA-PCA and CFA-PCA, achieve the computational minimax lower bounds (CMLBs) in different signal regimes, yielding a unified phase diagram for optimal clustering under block-structured signals. If the data do exhibit a contiguous signal structure, the proposed methods are more powerful than standard non-block methods in clustering and signal recovery, as demonstrated by our theoretical and numerical results. Such data sets are commonly seen in geophysical and climate applications. The MA-PCA and CFA-PCA procedures can adapt non-block signals as well by choosing zero window size ($h_1 = h_2 = 0$ and $h_3 = 1$). Our minimax result in Corollary 1 with $\alpha = 0$ shows that the non-block MA-PCA and CFA-PCA also achieve the CMLBs under the setting of non-block signals. Therefore, the proposed methods can accommodate both block and non-block signals by adaptively selecting the hyperparameters, indicating the benefits of applying the proposed method with

tuning window sizes.

In practical applications, the signal sparsity regime may be unknown. We recommend applying both CFA-PCA and MA-PCA, and then comparing the clustering results using post-clustering signal recovery and domain knowledge. Since signals for variable selection would be canceled if many labels from clustering are mismatched, the number of identified signals from post-clustering signal recovery can be used as a criterion for selecting clustering methods. Meanwhile, domain knowledge and scientific interpretability should also be considered in choosing the final clustering result, as indicated in the case study.

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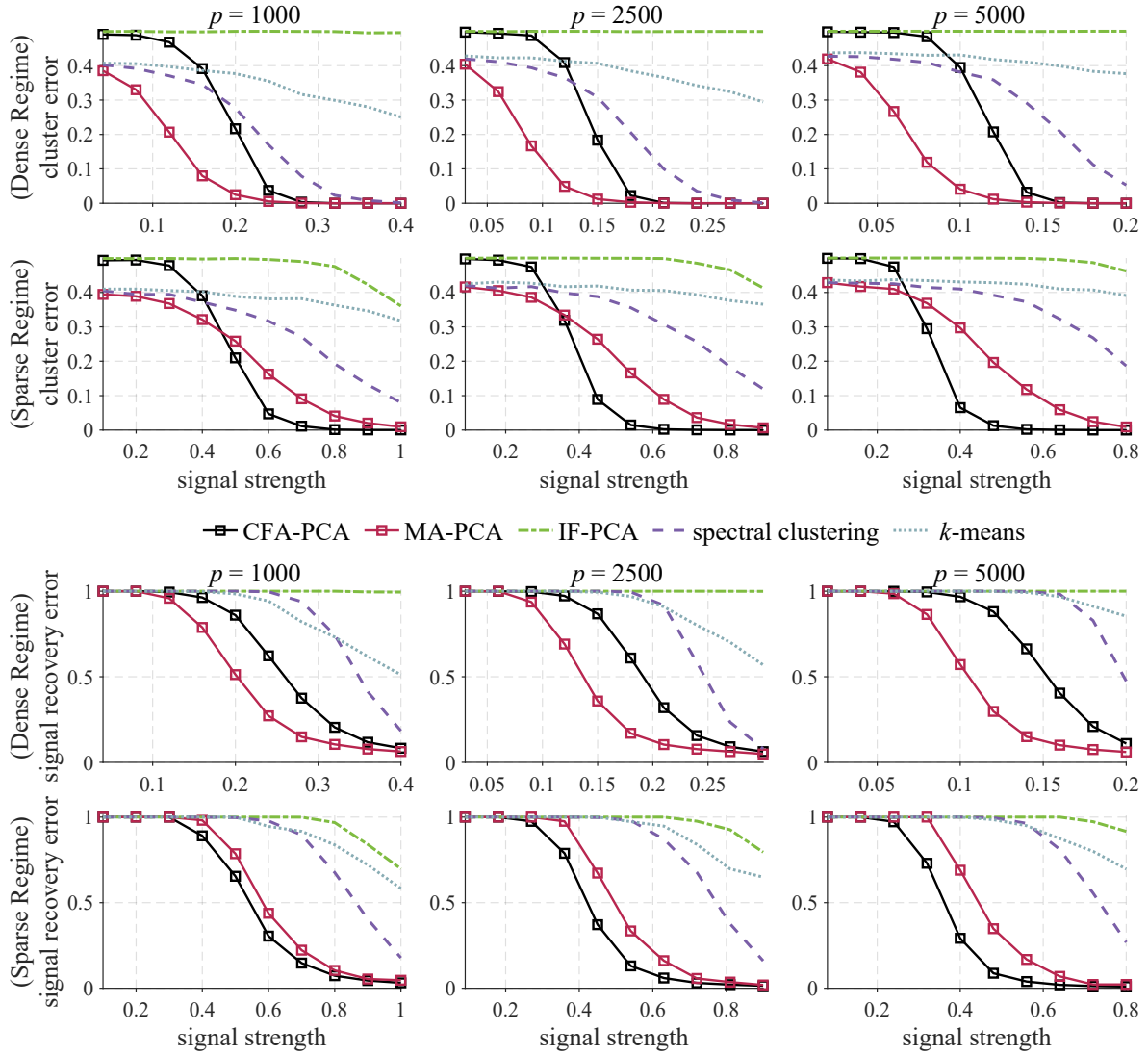


Figure 2: Simulation results for vector-valued data under block signal settings in Table S.1. The clustering errors (upper two panels) and signal recovery errors (bottom two panels) of CFA-PCA, MA-PCA, IF-PCA, spectral clustering and k -means under different signal strengths, averaged over 500 repeated experiments. The rows represent the dense and sparse signal scenarios and the columns represent different dimensions.